

Case Study Questionnaire

01 - Create Schema and Validate Tables

1. How can a dedicated schema be created for the e-commerce analysis project?
2. Which source tables were uploaded into Databricks?
3. How many records are available in each source table?
4. Were all required CSV files loaded successfully?
5. What are the column names, data types, and nullability details for each table?

02 - Database Exploration

1. What columns are available in the customer table?
2. What are the data types of all columns in the customer table?
3. What is the earliest order purchase date in the dataset?
4. What is the latest order purchase date in the dataset?
5. What is the total time range covered by the orders data in months and years?

03 - Order Trend Analysis

1. Is there a growing trend in the number of orders placed over the years?
2. How does order volume change month by month?
3. Is there any monthly seasonality in order placement?
4. During what time of day do customers mostly place orders?
 - Dawn: 0–6 hours
 - Morning: 7–12 hours
 - Afternoon: 13–18 hours
 - Night: 19–23 hours
5. What is the month-on-month number of orders placed in each state?

04 - Customer Geography Analysis

1. How many distinct customer cities and states are represented among customers who placed orders?
2. How are customers distributed across Brazilian states?
3. Which states have the highest customer concentration?
4. Which cities have the highest number of customers?
5. Is customer demand concentrated in a few regions or spread widely across Brazil?

05 - Economic Impact Analysis

1. What is the percentage increase in total payment value from 2017 to 2018, considering only January to August?
2. What is the total and average item price for each customer state?
3. What is the total and average freight value for each customer state?

4. Which states contribute the highest payment value?
5. Which regions appear to drive the strongest e-commerce money movement?

06 - Delivery and Freight Analysis

1. How many days did each order take to be delivered from the purchase date?
2. What is the difference between actual delivery date and estimated delivery date?
3. Which states have the highest average freight value?
4. Which states have the lowest average freight value?
5. Which states have the highest average delivery time?
6. Which states have the lowest average delivery time?
7. Which states receive orders fastest compared to the estimated delivery date?
8. Are deliveries generally early, late, or close to the estimated date?

07 - Payment Analysis

1. What payment types are used by customers?
2. How does monthly order volume vary across different payment types?
3. Which payment method is used most frequently?
4. Which payment method contributes the highest total payment value?
5. How are orders distributed by payment installments?
6. What does installment usage suggest about customer payment behavior?

08 - Business Insights Summary

1. What are the most important business observations from the analysis?
2. Which states should be prioritized for marketing, logistics, and seller support?
3. What do order trends suggest about e-commerce growth?
4. What do payment patterns suggest about customer preferences?
5. What do freight and delivery patterns suggest about logistics performance?
6. What actionable recommendations can be made for the business?